Underneath this, we find something more interesting: ants can find the food closest to them—meaning more chances of successfully transporting it home! How? It's simple: a given ant has a certain amount of pheromone inside it, and the ant that finds the closest food source can leave behind more of the pheromone. (The ant that finds food far away will have its pheromone exhausted by the time it gets home.) As a result, the colony gravitates towards not just any food, but towards what food is closest to them!

On the surface, the ants seem to be searching for food and finding it effectively. But backstage, it's all probability and pheromones. This isn't the whole story, but here's one way intelligence can come from simplicity.

Essentially, the ants "self-organise." Self-organisation refers to the fact that "good" patterns and configurations—as in the ants finding the nearest food—emerge from interactions between ants, and between the ants and the environment. These interactions must, naturally, be conducive to the emergence of intelligence: for example, Darwin would quickly kill off a species of ant where individuals simply lunched on the food they found.

We, Robots

Intelligence emerging from group activities is fine, but what about the individuals themselves? Leaving behind ants and/or bots for the moment, let's just call them "agents." So each agent's properties sheet would read something like the following.

Alignment: Each agent has the ability to align itself with the others: just find all the agents in the vicinity and average where they're headed. Then align yourself in that direction. Except for the first agent, who is headed god knows where, every other agent gets aligned because its neighbour is aligned because its neighbour...

Cohesion: An agent can approach, and form a group with, nearby agents. Find some of your neighbours, figure their "average" location, then move to that location.

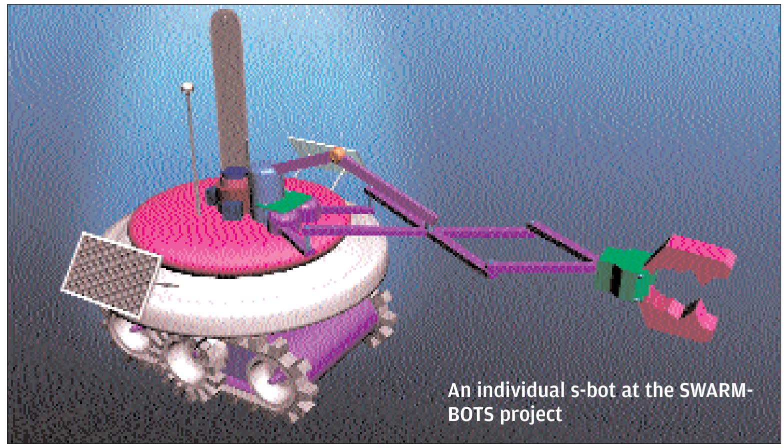
Obstacle avoidance: An obvious requisite. Have you ever seen an ant bump into a wall, then go back, and bump into the same wall again? Well, it does seem like ants have an inbuilt obstacle-avoidance algorithm, and any agent participating in a swarm needs one.

Separation: The ability to maintain a distance from nearby agents. This prevents crowding, which allows the agents to scout a wider area.

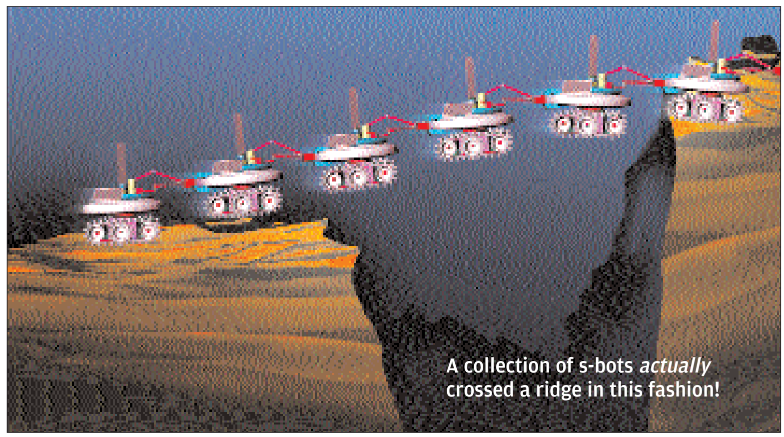
That's about behaviour intra-swarm. As for interactions with the environment, if we're talking about a little robot, it would need equipment that enables it to take *in* things from the environment—such as solar cells for power, sensors of various kinds, and more—and it needs to be built such that it can contribute *to* the environment: it needs grippers, wheels, and so forth.

Born To Bond

Summing up what we've said thus far: each robot needs to have some means and characteristics by which it can act as part of a swarm. Intelligence and useful behaviour can emerge from the swarm, and what kind of behaviour emerges depends on the properties of each member. The idea is to design little bots such that a swarm of them can do something useful. And various uses for robotic swarms have been envisaged, including terrestrial, outer-space and submarine exploration.



An individual s-bot at the SWARM-BOTS project



A collection of s-bots *actually* crossed a ridge in this fashion!

SWARM-BOTS was a project funded by the Future and Emerging Technologies programme of the European Community. Its focus was "self-organising, self-assembling, biologically-inspired robots." The project calls the swarm itself a "swarm-bot," which is an aggregate of s-bots. See illustration above: the little critters can self-assemble by connecting and disconnecting from each other using their claw-like grippers! The SWARM-BOTS project hoped to create groups that could "explore, navigate and transport heavy objects on rough terrains in situations in which a single s-bot would have problems to achieve the task alone."

Each s-bot itself has limited computational power, but apart from that, the spec-sheet of the s-bots is impressive. Each has "proximity sensors, light sensors, accelerometers, humidity sensors, sound sensors, an omni-directional colour camera, force sensors..." The swarm-bot as a whole was designed to reconfigure itself as and when needed: for example, falling into single file to go through a narrow passage. The project mandated that the swarm-bot aggregate should form as a result of the collective intelligence of the s-bots, rather than hard-coded rules.

Potential applications for the s-bots include space exploration, rescue searches, and underwater exploration. The project ended in March of last year; what came of it? Did the s-bots form swarms and explore Mars? Of course they didn't, because they were never sent there. One of the things that